

ROSEBANK COLLEGE

ASSESSMENT SUBMISSION: POE PART 1

RESEARCH, PLANNING, AND DESIGN SPECIFICATION

Module Name: Open-Source Coding (Introduction)

Module Code: OPSC6311

Project Name: BudgetQuest Personal Budget Tracker

STUDENT DETAILS

st10456349

st10445385

st10456794

st10446898

st10436179

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OPSC6311: POE Part 1 — Comprehensive Research and Design Specification

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1. Introduction and Project Scope

The management of personal finances is a foundational challenge faced by individuals across all income levels and demographics. In the specific context of South Africa, economic pressures such as high inflation, rising unemployment, and fluctuating exchange rates place immense strain on household budgets. This environment has created an urgent need for accessible, high-performance financial management tools that provide real-time awareness.

BudgetQuest has been developed to fulfill this need as an "offline-first" Android application. This project serves a twofold purpose:

1. To conduct a deep-dive comparative analysis of current market leaders.
2. To translate those research insights into a practical, innovative blueprint for a new application that solves common user pain points like "boring" manual entry and unreliable bank syncing.

2. Section 1: Academic Research and Comparative Analysis

To establish industry benchmarks and identify market gaps, three prominent Android applications were evaluated: **22Seven (Vault22)**, **Mint (by Intuit)**, and **YNAB (You Need A Budget)**.

2.1 22Seven (Vault22)

22Seven is a leading South African application that focuses on account aggregation. By linking directly to local financial institutions, it automates the process of data collection.

- **Strengths:** The automated categorization of transactions reduces the "friction" of manual entry, making it highly convenient for busy professionals.
- **Weaknesses:** User research indicates significant frustration regarding "broken syncs." When bank APIs fail, the user is left with an incomplete financial picture. Furthermore, the auto-categorization often mislabels transactions, requiring tedious manual corrections.

- **Innovative Element:** Its "Nudges" feature provides personalized insights into spending habits, such as identifying recurring subscriptions that may no longer be necessary.

2.2 Mint (by Intuit)

Mint is widely considered the global standard for UI/UX in the personal finance space. It prioritizes visual storytelling through a variety of charts and graphs.

- **Strengths:** Mint excels at providing a "big picture" view of a user's net worth, including investments, properties, and debts. Its interface is clean, intuitive, and highly responsive.
- **Weaknesses:** As a US-centric application, its support for South African banks is non-existent. Additionally, its focus on "past spending" rather than "future planning" makes it less effective for users trying to break the paycheck-to-paycheck cycle.
- **Innovative Elements:** One of Mint's most innovative contributions to personal finance was the integration of free credit score monitoring within a budgeting application. At the time, accessing a credit score typically required a separate paid service. By combining credit monitoring with budgeting tools, Mint provided users with a more holistic view of their financial health, linking spending behavior directly to credit outcomes. This integration remains relatively uncommon in free budgeting apps and highlights Mint's forward-thinking approach.

Another key innovation was its subscription tracking feature. Mint automatically identified recurring payments, alerted users to price increases, and enabled easy review of ongoing expenses. This addressed the growing challenge of managing multiple subscription services, helping users reduce unnecessary spending and maintain better control over their finances.

These features demonstrate Mint's emphasis on financial awareness and automation, offering practical solutions to real-world financial management challenges.

2.3 YNAB (You Need A Budget)

YNAB follows a strict "Zero-Based Budgeting" philosophy, where every Rand is assigned a specific task before it is spent.

- **Strengths:** It is exceptionally effective at changing user behavior. It treats a budget as a living document that must be adjusted whenever circumstances change.
- **Weaknesses:** The learning curve is steep. Many new users find the methodology overwhelming and eventually abandon the app due to the high level of manual maintenance required.
- **Innovative Elements:** YNAB's most groundbreaking feature is how it incorporates a full financial philosophy into the app experience instead of just being a tool. The "Age Your Money" metric is a truly unique idea in consumer finance apps: it measures the days between when you earn money and when you spend it, motivating users to slowly create a buffer by using "older" money. As this metric gets better over time, users build a real financial safety net that shields them from unexpected expenses — changing the app from just a record-keeping tool into a real agent of behavioral change.

Another innovative conceptual feature is "Embrace your true expenses." While many budgeting apps view budgeting as a monthly task, YNAB's method encourages users to divide large, infrequent expenses (like yearly insurance payments, car maintenance, or holiday expenses) into smaller monthly amounts. This helps avoid the "budget shock" that happens when users overlook irregular expenses and find themselves unable to pay for them. This feature fills a crucial gap in how most people perceive budgeting.

2.4 Research Conclusion

The synthesis of this research reveals a clear market opportunity. While existing apps are functional, they lack the "fun" element required to keep users motivated over months and years. There is a total absence of gamification in the current landscape. BudgetQuest is designed to fill this void by treating financial health like a character-building quest in a game.

Comparative Analysis of Budgeting Application Features:

The table below provides a structured infographic-style comparison of the three apps across eleven key evaluation dimensions. Ratings are provided on a five-point scale for quick visual reference, supplemented by contextual notes.

Dimension	22Seven / Vault22	Mint (Intuit)	YNAB
Pricing	Free (ZA only)	Free (was ad-supported)	\$14.99/mo or \$109/yr
Platform	Android & iOS	Android & iOS (shut down)	Android, iOS, Web, Apple Watch
Account Linking	120+ SA institutions	US/global banks & cards	Global banks (Plaid/MX)
Budgeting Method	Category-based tracking	Automated tracking	Zero-based (ZBB)
Automation Level	High (auto-import)	High (auto-import)	Medium (manual + auto)
Learning Curve	Low	Low	High (2–4 months)
Gamification	Financial Fitness Levels	None	None
Reporting & Graphs	Moderate	Good (pie charts, history)	Excellent (detailed)
Alerts & Notifications	Budget alerts, Nudges	Alerts, bill reminders	Category overage alerts
Unique Feature	SA bank integration + Nudges	Credit score monitoring	Age Your Money metric
Current Status	Active (rebranded Vault22)	Shut down Mar 2024	Active

3. Section 2: Planning and Design Specifications

3.1 App Overview and Branding

Application Name: BudgetQuest

App Icon Concept: The app icon uses a gold coin and upward arrow to represent financial growth and smart budgeting. The green background symbolizes wealth and stability, while the clean design ensures the app is visually appealing and easy to recognize.



3.2 Innovative Features and Gamification Logic

BudgetQuest differentiates itself through three primary innovations:

1. **"Money Milestones" (XP System):** To drive engagement, the app implements an Experience Point (XP) system. Users earn XP for positive financial behaviors.
 - a. +10 XP: Logging an expense.
 - b. +25 XP: 3-day logging streak.
 - c. +50 XP: Finishing a month under budget.
2. **XP Levels:** Users progress through ranks:
 - a. *Penny Pincher* (0–199 XP)

- b. *Budget Apprentice* (200–499 XP)
 - c. *Money Manager* (500–999 XP)
 - d. *Finance Sage* (1,000–1,999 XP)
 - e. *BudgetQuest Master* (2,000+ XP)
3. **Receipt Attachment:** Addressing a gap in competitors, BudgetQuest allows users to capture a photo of a physical receipt and attach it to an expense. This ensures that manual entries are verifiable and organized.

3.3 Functional and Non-Functional Requirements

Functional Requirements:

- **User Management:** Secure local registration and login.
- **Transaction Logging:** Create, edit, and delete expenses with amount, date, and category.
- **Budgeting:** Set global monthly spending goals and individual category limits.
- **Visualization:** Generate pie charts and bar graphs based on historical data.

Non-Functional Requirements:

- **Offline-First:** All data must be stored locally using a RoomDB database to ensure privacy and speed.
- **Performance:** The app must target API Level 26 (Android 8.0) and above, ensuring compatibility with the majority of modern Android devices.

4. Section 3: User Interface (UI) Design

The UI design focuses on "glanceability"—the ability for a user to understand their financial status in under five seconds.

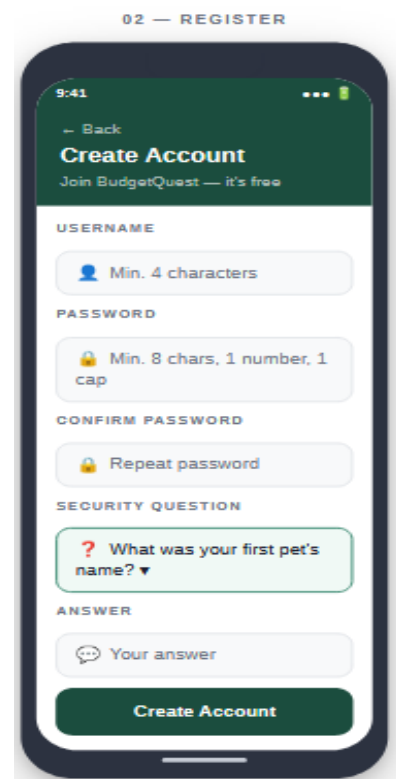
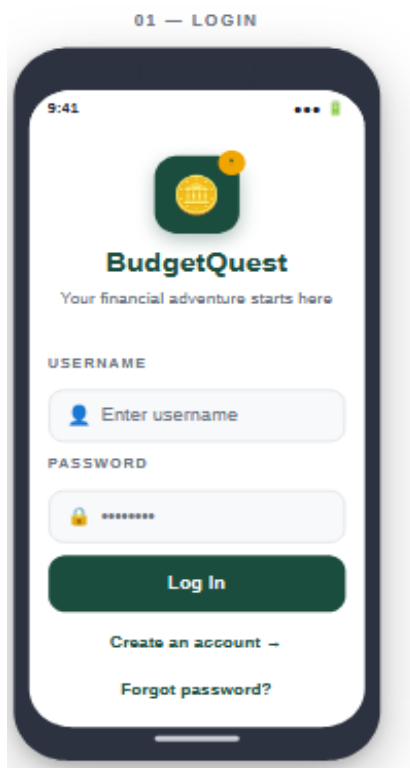
4.1 Navigation Strategy

The app utilizes a **5-Tab Bottom Navigation Bar** providing direct access to:

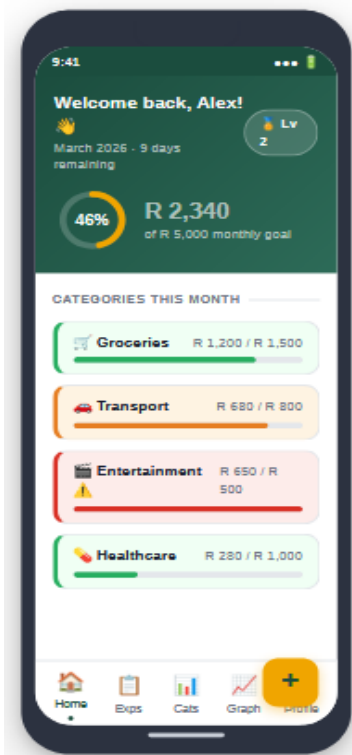
- **Dashboard:** High-level summary and XP status.
- **Expenses:** Detailed transaction list.
- **Add Expense:** Quick-entry form (via Floating Action Button).
- **Graphs:** Visual data analysis.
- **Profile:** Gamification hub and settings.

4.2 UI Wireframes and Mockups

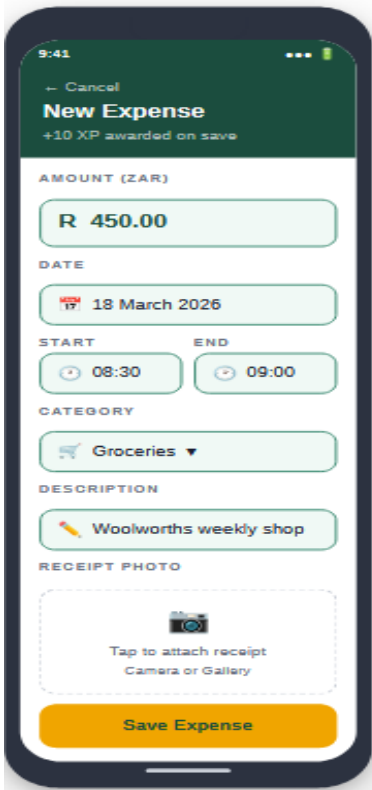
The following screens represent the core user flow:



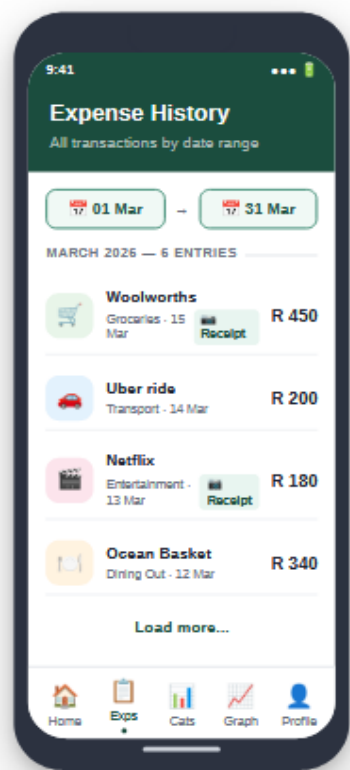
03 — DASHBOARD (HOME HUB)

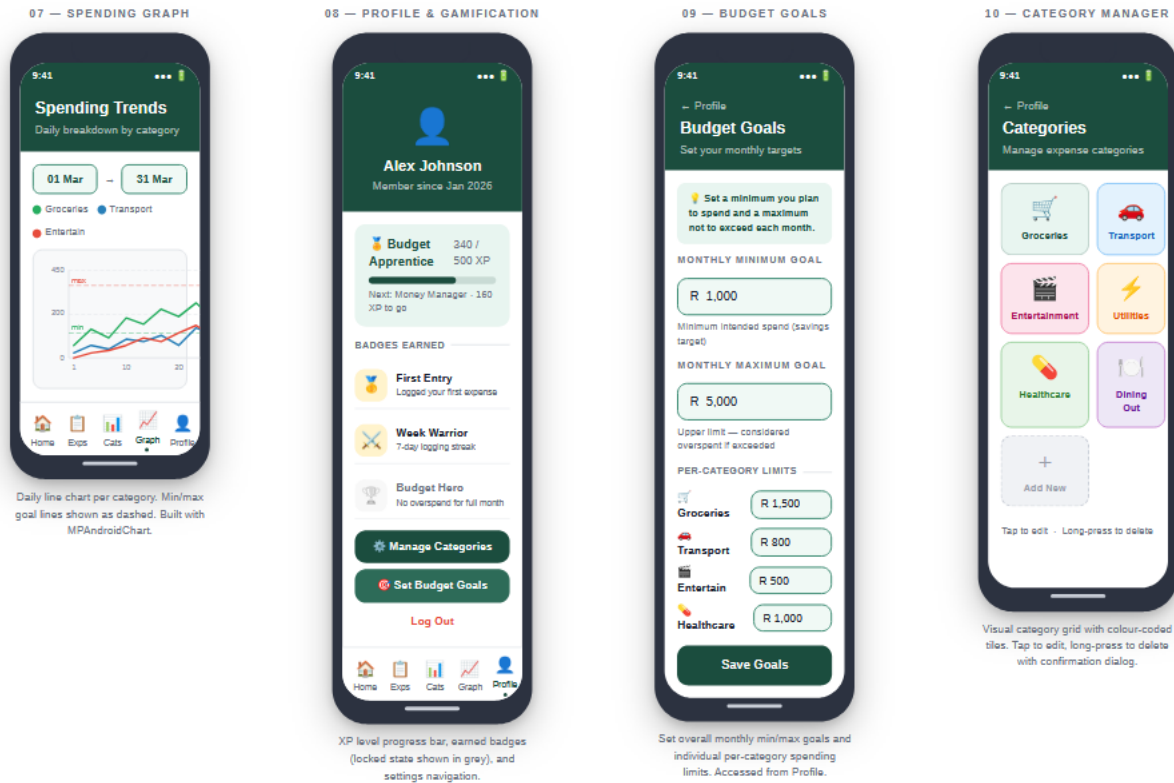


04 — ADD / EDIT EXPENSE



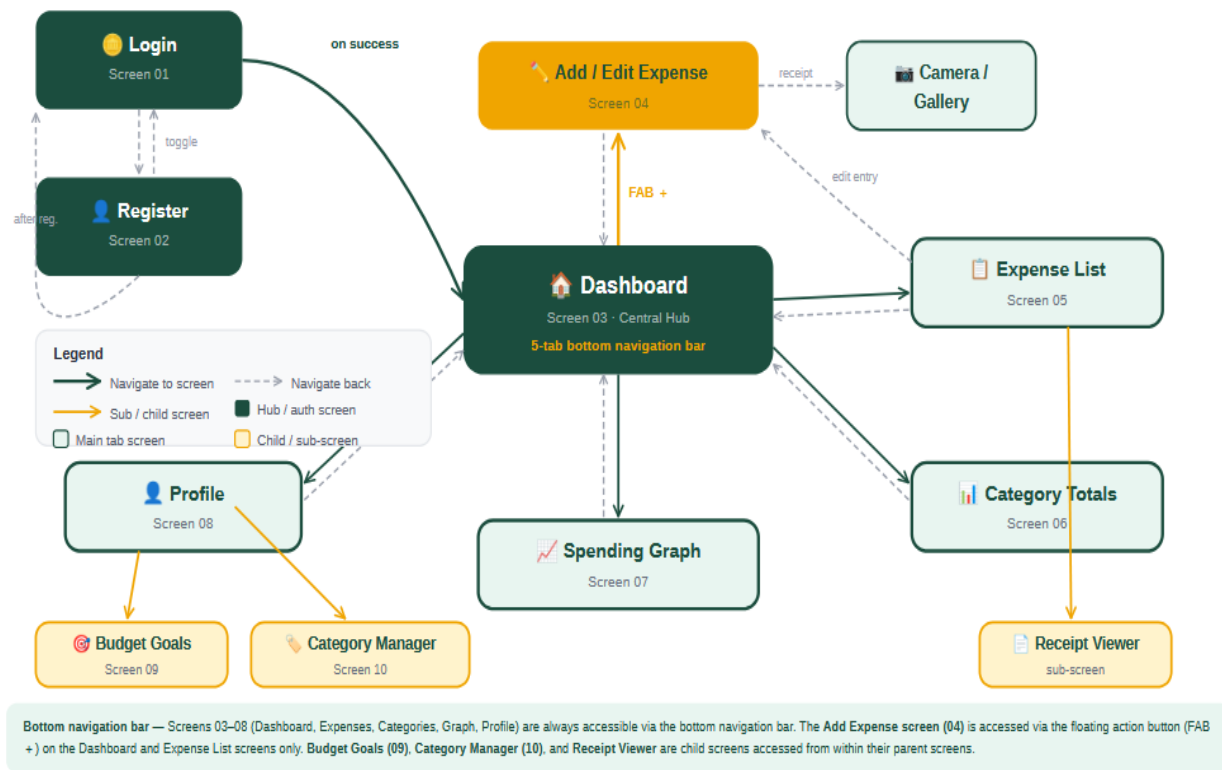
05 — EXPENSE LIST





4.3 Navigation Flow Diagram

The routing between these screens is designed to be circular, ensuring the user can return to the Dashboard from any sub-menu with a single tap or "Back" gesture.



5. Section 4: Project Management Plan (Gantt Chart)

The development of BudgetQuest follows a structured 10-week timeline, prioritizing the core database architecture before layering on the advanced gamification features.

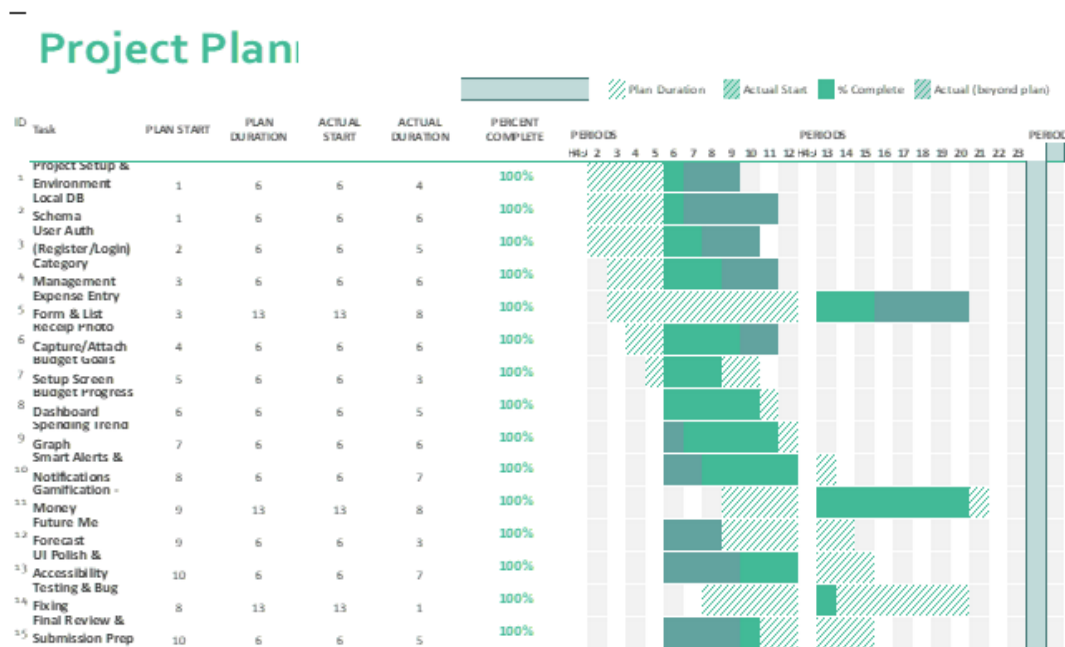
5.1 Project Methodology

We are utilizing an **Agile-Lite** methodology. This allows for iterative testing of the RoomDB database schema before the UI is fully finalized.

- **Weeks 1-2:** Requirement analysis and Research Finalization.
- **Weeks 3-5:** Database architecture and User Authentication.
- **Weeks 6-8:** Camera API integration and Gamification logic.
- **Weeks 9-10:** Final QA, bug fixing, and documentation.

5.2 The Gantt Chart

The following chart outlines the specific tasks, durations, and milestones for the project.



6. Conclusion

BudgetQuest is more than a standard expense tracker; it is a behavioral change tool. By addressing the "boredom" of manual tracking through an XP-based rewards system and solving the "reliability" issue with a local-first RoomDB approach, this application is positioned to provide superior value to South African users. The planning and design documented here provide a robust foundation for the implementation phase in Android Studio.

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